IATSS Research 49 (2025) 481–492

Contents lists available at ScienceDirect

IATSS Research

journal homepage: www.sciencedirect.com/journal/iatss-research

Research Article

Adaptive traffic signal control using deep reinforcement learning: A multi-objective approach for single and multi-intersection scenarios

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ARTICLE INFO

Keywords:
Adaptive traffic signal control (ATSC)
Deep reinforcement learning (DRL)
Traffic safety
Mobility
Carbon dioxide emissions
SUMO

ABSTRACT

Traffic congestion in urban networks necessitates adaptive signal control systems that balance efficiency, safety, and environmental impact. While Deep Reinforcement Learning (DRL) has shown promise for traffic signal control (TSC), existing approaches often optimize single objectives or lack scalability for multi-intersection corridors. This study proposes a multi-objective DRL framework that simultaneously minimizes vehicle delays, traffic conflicts (using Time-to-Collision metrics), and CO_2 emissions. Six reward functions were designed and tested with three DRL algorithms (PPO, A2C, DQN) in synthetic and real-world intersections simulated in SUMO. Key results demonstrate that: (1) PPO outperformed A2C and DQN, achieving up to 21 % fewer conflicts and 7 % lower delays in high-traffic scenarios; (2) A safety-focused reward (Reward 3) reduced conflicts by 4–20 % compared to fixed-time controls in high-traffic scenarios, while multi-objective rewards (Rewards 5–6) balanced all targets effectively; and (3) Decentralized control for multi-intersection corridor reduced delays, conflicts, and emissions by 26.4 %, 26.9 %, and 12 %, respectively, surpassing centralized approaches. Real-world validation in Hangzhou, China, and Cologne, Germany, confirmed robustness, with 10–15 % delay reductions and 2.2–6.4 % lower emissions versus prior DRL models. This work advances adaptive TSC by integrating safety and sustainability into DRL optimization, offering scalable solutions for heterogeneous traffic networks.

1. Introduction

Traffic congestion is a major issue in urban settings, causing longer travel durations, increased fuel consumption, and more environmental pollution. [1]. To address this problem, implementing a more efficient traffic signal control system proves to be an effective solution [2]. Traffic signal control is a key feature of an intelligent traffic system and a crucial approach to resolve traffic challenges. [3].

Adaptive Traffic Signal Control (ATSC), a critical component of Intelligent Transportation Systems, dynamically optimizes signal timing based on real-time data, in accordance with a defined objective function, and takes into account both real-time traffic data and the prediction of possible changes in the traffic flow [4]. ATSC controllers can handle fluctuations in traffic volumes more effectively, allowing them to improve the overall operational capacity and efficiency of a single intersection or an entire network of connected intersections [5].Several adaptive signal control systems have been developed accordingly.

In recent years, the fields of machine learning and artificial intelligence, specifically deep reinforcement learning (DRL), have shown great potential for generating, evaluating, and enhancing decision-making solutions for adaptive traffic signal control [6,7]. DRL combines the powerful pattern recognition capabilities of deep neural networks with the optimization principles of reinforcement learning, enabling the development of adaptive signal control policies that can learn to optimize traffic flow directly from large-scale traffic data [8].

Despite notable progress, most existing DRL-based ATSC models focus predominantly on improving traffic efficiency, while giving limited attention to safety and environmental sustainability. For instance, Genders and Razavi [9], Pan [10], and Guo et al. [11] prioritized mobility metrics such as delay and throughput, without incorporating conflict or emission objectives. Some recent works have extended toward multi-objective control, such as Gong et al. [12], who incorporated crash risk, and Schumacher et al. [13], who considered CO_2 emissions. However, these efforts were constrained to isolated intersections and typically relied on a single DRL algorithm, reducing flexibility and generalizability. To bridge these gaps, this study proposes a multi-objective DRL framework that simultaneously optimizes traffic flow, safety, and emissions across diverse infrastructures—from isolated

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https://doi.org/10.1016/j.iatssr.2025.10.004
Received 19 January 2025; Received in revised form 1 September 2025; Accepted 13 October 2025
Available online 24 October 2025
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intersections to large-scale urban networks. By leveraging three state-of-the-art DRL algorithms (PPO, A2C, and DQN), the framework supports comprehensive trade-off analysis among conflicting objectives and enables scalable, adaptable, and policy-driven traffic control solutions that promote smarter, greener, and safer urban mobility.

2. Literature review

Over the past years, adaptive traffic light control systems have gained considerable interest, especially with the application of (DRL) [14]. Research efforts to enhance ATSC using DRL often focus on one or a combination of these three key areas, namely, (1) state representation, (2) reward definition, and (3) action space definition [15]. To contextualize these advances, we first contrast DRL with classical rule-based and non-DRL machine learning approaches before detailing DRL-specific components.

2.1. Comparison with traditional and machine learning methods

Traffic signal control strategies can be broadly categorized into three traditional types. The first is pre-timed control, which assigns fixed green phase durations based on historical traffic data, irrespective of real-time traffic fluctuations. The second is vehicle-actuated control, which relies on traffic demand data from detectors (e.g., inductive loops) to dynamically extend or terminate signal phases [1,16]. The third is adaptive traffic signal control (ATSC), which, like actuated control, responds to real-time traffic data but with more complex algorithms that optimize timing strategies based on defined objective functions [17].

Adaptive systems optimize signal timings network-wide using real-time flow measurements. However, these methods rely on predefined control logics and extensive infrastructure and often struggle to manage nonlinear traffic patterns or multiple competing objectives. Overall, while these rule-based and adaptive systems are effective under stable conditions, they operate under rigid assumptions and typically optimize a single performance criterion, such as delay [4].

To overcome these limitations, machine learning (ML) approaches have been explored. Supervised learning methods (e.g., decision trees, neural networks) require labeled data, which are scarce in traffic signal control since historical plans are rarely evaluated for optimality [18]. Unsupervised methods such as clustering have been applied to recognize traffic patterns or manage uncertainty. However, these methods lack the ability to adaptively improve control policies through interaction with the environment.

More recently, various artificial intelligence (AI) techniques have been applied to traffic signal control, including fuzzy logic, swarm intelligence, and reinforcement learning (RL). Among these, RL has gained significant attention due to its ability to learn through trial and error, making it well-suited for complex, real-time, and multi-objective control problems.

In particular, deep reinforcement learning (DRL) enables agents to learn optimal traffic signal policies by interacting with the environment, receiving feedback (e.g., delays, emissions, conflicts), and refining their decisions over time, without requiring labeled datasets. This combination of adaptability, sequential learning, and multi-objective optimization makes DRL a highly effective and flexible solution compared to both traditional rule-based systems and non-RL machine learning approaches [18].

2.2. State representation

In DRL-based TSC research, the state encompasses the agent's perception of traffic intersections, which informs its decision-making process at each time step. The state is generally the combinations of: Queue length, Waiting time, Volume (number of vehicles on the lane/road), Speed, Phase duration of the current phase, Congestion level denoted by 0/1.

Genders and Razavi, 2016 [9] introduced the concept of discrete traffic state encoding (DTSE) that has been widely used in defining traffic states since then. DTSE divides each lane into cell segments, gathering pertinent information for an approaching vehicle per cell. Formally, DTSE comprises three components: a vector indicating the presence or absence of a vehicle, a vector detailing the vehicle's speed, and a vector showing the current traffic phase. Other researchers [19] suggested using a virtual snapshot image of the current intersection obtained from a simulator and dividing it into a grid of cells. Each cell contains relevant information, such as the vehicle's position and speed.

2.3. Action formulation

The action generally has four types: adjusting the current phase duration, setting phase ratios, deciding to continue or switch phases, and selecting the next phase.

2.4. Reward formulation

In the context of deep reinforcement learning-based traffic signal control, the reward function represents the learning objective of the system. This reward typically encompasses the following key elements: queue length, waiting time, cumulative delay, speed, throughput and pressure, which is the difference between the number of incoming vehicles and the number of outgoing vehicles at the intersection. Some research works used a weighted linear combination of the above items [10,12,20,21].

2.5. Recent studies on DRL for TSC

Many studies highlight DRL's superiority in traffic optimization. Table 1 summarizes studies using DRL for traffic signal control. However, most focus solely on traffic efficiency, neglecting safety and environmental factors. For example, studies like [9,10,17,22,23] do not address safety or emission-related objectives.

Some research has tackled broader goals. For instance, [12] employed multi-objective reinforcement learning with a real-time crash risk model to balance safety and efficiency, while [24] extended to safety, efficiency, and decarbonization. However, these efforts were limited to isolated intersections and relied solely on the Double Dueling Deep Q-Network (DQN3) algorithm, reducing methodological variety.

This study bridges the gap by proposing an adaptive traffic signal controller with a range of objectives including mobility, safety, CO_2 emissions, as well as multi objective optimization that integrates safety and CO_2 emissions alongside mobility in a multi-objective optimization framework. Unlike previous work limited to isolated intersections, this study addresses both single and multi-intersection scenarios. It employs three DRL algorithms—Proximal Policy Optimization (PPO), Advantage Actor-Critic (A2C), and Deep Q Network (DQN)—to comprehensively analyze trade-offs among mobility, safety, and environmental objectives.

The primary objective of this study is to develop an adaptive traffic signal controller that balances mobility, safety, CO_2 emissions, and multi-objective optimization for single and multi-intersection setups. Its effectiveness will be evaluated against traditional fixed and actuated signal controls.

The remainder of this paper is structured as follows: Section 3 provides theoretical background on DRL in TSC; Section 4 outlines experiment design, including state representation, action formulation, and reward functions; Section 5 presents results comparing DRL-based and traditional methods; and Section 6 concludes the research.

3. Background

This section provides an overview of the fundamental concepts and

Table 1
Some of the recent studies on DRL to develop traffic signal controller for single intersection.

Reference	State	Reward	Simulator	Compared Methods	Best Method
Han et al. [20]	- DTSE	- Throughputs, - Pedestrian waiting time, - Exceeding pedestrian maximum red time, - Exceeding minimum green time.	SUMO	- Fixed time signal - Dueling DQN - Improved Dueling DQN	Improved Dueling DQN
Pan [10]	- Queue length - Number of vehicles - Total waiting - Current Phase - Vehicles positions.	- Total delays - Total waiting time - Total queue length - Change of phase - Vehicles passed the intersection - Total travel time	SUMO	- Fixed time signal - DQN	DQN
Guo et al. [11]	- Queue length	- Difference in queue length between time step	SUMO	- DQN - Fixed-time control - Gap-based control - Time loss-based control	DQN
Huang & Qu [25]	- Traffic volume - Average vehicle speed - Average queue length - Average waiting time - Lane occupancy	- Average speed	SUMO	- PPO - Actuated method - SCOOT	PPO
Li et al. [26]	-Speed - Queue length - Density	- Traffic delay - Queue length - Change in density of all roads between states	PARAMICS	- DQN - Conventional RL	DQN
Jörneskog and Kandelan [27]	- Mean speed of vehicles and pedestrians for each incoming road	- Change in the mean speed of all roads between states	SUMO	- DQN - Fixed - Deterministic	DQN for uneven distribution of vehicles
Genders and Razavi [9]	- DTSE	- Change in cumulative vehicle delay between actions.	SUMO	- DQN - Conventional QL	DQN
Gong et al. [12]	- Location of vehicles - Current activated signal phase	- Difference between current and previous cumulative waiting time. - Risk score using a crash risk model based on historical crash and traffic data.	Aimsun	- Multi objective DQN3 (optimizes traffic efficiency and safety) - Single objective DQN3 (traffic efficiency only) - Actuated signal controller	- Multi objective - DQN3
Wu et al. [28]	- State1: Current signal time and total number of vehicles on each lane - State2: Position and speed	- Queue length - Waiting time - Throughput - Change in the number of approaching vehicles on the lane between states	VISSIM	- DQN - A2C - Dueling DQN - Double DQN - Fixed - ASC	- Dueling DQN in reducing delay - ASC in reducing number of stops
Schumacher et al. [13]	- DTSE	- queuing length - speed - braking acceleration - CO2 emission rates - Queue + brake	SUMO	- Compared different rewards using DQN	- The queue + brake reward was effective in reducing CO2 emissions, and in both traffic scenarios. - The speed reward achieved low emissions and high speed

The Action: Update the next signal phase either continuing using the current phase, using pre-defined set of phases, or try new phase.
DTSE: Discrete Traffic State Encoding.
PPO: Proximal Policy Optimization.
DQN: Deep Q Network.
DQN3: Double Dueling Deep Q Network.
ASC: Actuated signal controller.
QL: Q Learning.

theoretical foundations in the fields of reinforcement learning and deep reinforcement learning that are relevant to this research.

3.1. Reinforcement learning

Solving ATSC problem under RL can be modeled as a Markov Decision Process (MDP) [29]. At each time step, the controller observes the current state S_t and selects an action A_t to execute. This causes the system to transition from state S_t to the next state S_{t+1}, and the controller receives a real-valued reward R_t. All these records, the state (S_t), action (A_t), reward (R_t), and next state (S_{t+1}) are stored, with the goal of finding the best policy that maximizes the total reward. The expected cumulative discounted future reward at a state is defined as the value function of the state *V(s)* or state–action pair *Q(s, a)*, also called Q-value, and can be calculated as:

$$Q(s_t, a_t) = E(R_t|s_t, a_t) = E\left[\sum_{i=1}^{\infty} \gamma^i r_{i+1}\right] \quad (1)$$

Where γ is the discount factor a value between 0 and 1 that determines the importance of future rewards.

The controller seeks to achieve the optimal policy, π^*, which can be determined by selecting the highest Q value for the given state "s" as follows:

$$\pi^* = \text{argmax}Q(s, a) \quad (2)$$

3.2. RL algorithms

RL traffic signal control measures can be divided into three types, namely: (1) Value-based methods; (2) Policy-based methods; and (3)

Actor-Critic methods.

Value-based methods estimate either the state-value function or the state-action value, and the policy is derived indirectly from the learned value function. These methods are limited to discrete actions because each action requires its own separate modeling process [18].

The optimal Q-value function is defined by the Bellman Eq. [30] and this function can be learned through an iterative update process as:

$$Q_{t+1}(s,a) = Q_t(s,a) + \alpha(r_{t+1} + \gamma \max Q_{t+1}(s,a) - Q_t(s,a)) \quad (3)$$

where $0 < \alpha < 1$ is the learning rate. The Q-values aid the agent in deciding which action is most desirable to select in a certain state. When selecting an action, the agent chooses between exploration and exploitation. In exploration, a random action is chosen to update the Q-value., while exploitation selects the best-known action to maximize the state value [31].

The policy-based methods aim to directly learn the probability of taking different actions in each state. Unlike value-based RL methods that build the control policy step-by-step, policy-based methods start with a complete policy and refine it over time. Instead of receiving rewards immediately after each action, the agent receives rewards at the end of an episode, which could be a complete traffic simulation involving multiple signal changes.

In the Actor-Critic methods, the actor is a policy-based component that selects the best action according to its current policy. Simultaneously, Critic is a value-based component that gets the reward from the environment.

3.3. Deep reinforcement leaning

In traditional approaches, the Q-function is typically constructed using either a table or a function estimator. However, in traffic signal timing problems, the state spaces are so vast that solving the associated reinforcement learning problem within a reasonable time frame becomes challenging with table-based Q-learning. Moreover, traditional Q-learning methods using function approximators often struggle to capture the complex dynamics of traffic flow. In contrast, deep reinforcement learning (DRL) leverages deep neural networks as function approximators, enabling the management of complex environments with large state spaces, such as those involving images or sensor data [26].

4. Experiments design

In this section, we perform experiments using both synthetic and real traffic data for a single intersection.

4.1. Environment setting

This study examines traffic signal control for both single and multi-intersections. For single intersections, we consider synthetic and real scenarios. The synthetic setup includes a four-way intersection with two incoming and outgoing lanes per direction, 300 m long, with a speed limit of 13.89 m/s and a lane width of 3.65 m. The geometry is depicted in Fig. 1**.a.**Traffic cases include balanced and unbalanced scenarios as outlined in Table 2, with flow ratios of 2:1 between directions and 80 % through, 10 % left, and 10 % right turns. Traffic volumes range from low to high, determined using Synchro software [32]. To adapt DRL models for Egyptian intersections, we calibrated the Wiedemann-99 car-following model parameters from [33].

Two real intersections were also used in the study. The first is a simple, four-way intersection located in Hangzhou City, China, with four directions and two lanes each. The right lane is designated for through traffic, and the left lane is reserved for left turns, as shown in Fig. 1**.b**. The traffic flow data covers the morning peak hour from 7:00 AM to 8:00 AM, provided by [34]. The second intersection is more complex, as shown in Fig. *1*.**c**, and is based on a well-established SUMO scenario known as "TAPAS Cologne" [35]. This scenario models traffic within the real-world city of Cologne, Germany, including road network layout and calibrated demand data.

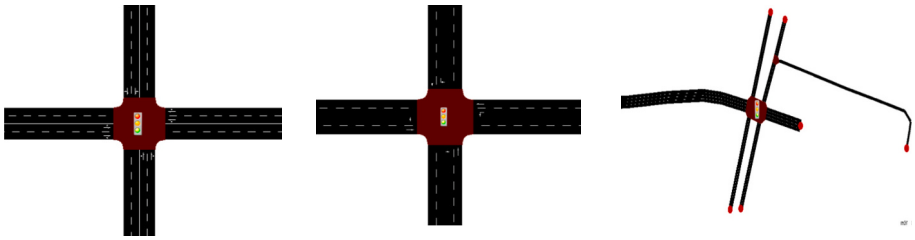
Expanding beyond single-intersection control, this study applies (DRL) to coordinate control of multiple intersections along an arterial corridor using the "TAPAS Cologne" [35] SUMO scenario as shown in Fig. 2.

4.2. Simulation setting

The experiments were conducted using the Simulation of Urban Mobility (SUMO) Version 1.18 [36]. SUMO is an open-source traffic simulation program that supports various vehicle types, multi-lane roads, traffic lights, and more. The interaction between SUMO and the proposed algorithm was facilitated through the Traffic Control Interface (TraCI) Python module. TraCI retrieves traffic information from the simulated environment and transfers it to the algorithm, which then determines t.

Table 2
Traffic scenarios for synthetic intersection.

Traffic Type	Scenario	Total Traffic volume (veh/h)	v/c ratio (synchro)	Car following model
Balanced	Scenario 1	2000	0.37	Wiedmann 99 (cc0 = 2, cc1 = 1.99, cc2 = 7.86, cc3 = −7.11, cc4 = −0.35, cc5 = 0.35, cc6 = 11.14, cc7 = 0.25, cc8 = 3.5, cc9 = 1.5)
	Scenario 2	2800	0.54	
	Scenario 3	3600	0.8	
	Scenario 4	4000	0.95	
Unbalanced	Scenario 5	1800	0.34	
	Scenario 6	2400	0.47	



(a) (b) (c)

Fig. 1. The Intersections geometry used throughout this Study.

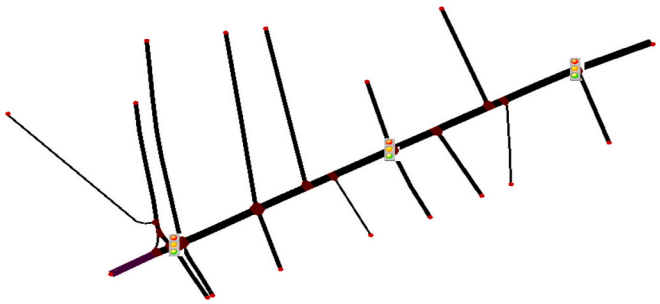


Fig. 2. Multi-intersections geometry used in this study.

4.3. Model formulation

The main elements of the proposed framework consist of the state formulation, reward function, and action space. The following sections offer a detailed explanation of each component.

4.3.1. State formulation

In this study, the state is defined using one of the following alternatives:

- **State 1:** A vector indicating the current active green phase and density of vehicles in each lane. In our implementation, density is calculated as the ratio of the number of vehicles on a lane to the maximum vehicle capacity of that lane. This approach allows for a normalized measure of congestion, ensuring consistency in representation across different traffic scenarios.
- **State 2:** A vector indicating the current active green phase and the queue length for each lane (with a vehicle considered queued if its speed is below 0.1 m/s).
- **State 3:** A vector indicating the current active green phase, the density for each lane, and the queue length for each lane.
- **State 4:** A vector representing the current active green phase and a position matrix encoding the presence or absence of vehicles at each intersection, as well as their speed.

4.3.2. Action formulation

The action space is discrete. Every 'delta_time', each traffic signal agent can choose the next green phase configuration. There are two restrictions on action selection: the agent can only choose to change the phase if the current phase duration exceeds the minimum green time and can only maintain the current phase if the duration is less than the maximum green time. Additionally, when a phase change occurs, a yellow phase with a fixed duration of 4 s is imposed.

4.3.3. Reward design

In this study, we designed six reward functions, each targeting specific aspects of traffic management:

Reward 1: This reward represents **traffic efficiency** by calculating total delays experienced by all approaching vehicles at each time step.

Reward 2: Also focusing on **traffic efficiency**, this reward measures total number of stopped vehicles at each time step.

Reward 3: This reward addresses **traffic safety** by accounting for total number of conflicts at each time step, serving as a surrogate measure of safety. We utilize SUMO software's capability to output simulated traffic conflicts by equipping all vehicles with a Surrogate Safety Measure (SSM) device. Specifically, we use **Time-to-Collision (TTC)** as the SSM in this research. TTC is defined as the expected time for two vehicles to collide if they continue at their current speed and trajectory [37]. A lower TTC value indicates a higher severity of a potential crash. Following the guidelines from [37], we consider a collision imminent if the TTC is less than 1.5 s. Therefore, we set the TTC threshold value to **1.5 s** in our experiments.

Reward 4: This reward focuses on Carbon dioxide emissions, the CO_2 emissions in this study were calculated using SUMO platform, which incorporates the HBEFA (Handbook Emission Factors for Road Transport) [38] model. SUMO provides vehicle emission data based on vehicle type, speed, acceleration, and driving patterns, and outputs the corresponding CO_2 emissions for each time step of the simulation. CO_2 was primary focus in this study due to its strong correlation with fuel consumption and its role as a major greenhouse gas [39].

Reward 5: A **multi-objective reward** that combines three traffic objectives with equal weights:

- 0.33 × Reward 1 (efficiency)
- 0.33 × Reward 3 (safety)
- 0.33 × Reward 4 (CO_2 emissions)

Equal weights were chosen to ensure a balanced consideration of all three objectives, efficiency, safety, and CO_2 emissions, since each is crucial for effective traffic management. This approach avoids bias toward any single objective. These weights can be established based on government policies and can be adjusted as necessary to adapt to specific situations. **Reward 6:** This **multi-objective reward** balances two critical traffic objectives, efficiency and safety, by assigning equal weights to each. This approach ensures that both aspects are given equal consideration, reflecting their importance in effective traffic management. The weights are distributed as follows:

- 0.5 × Reward 1 (efficiency)
- 0.5 × Reward 3 (safety)

4.4. Model setting

All algorithms trained in this study were implemented using deep neural networks as function approximators. In this study, we employ three popular DRL algorithms: Deep Q-Network (DQN), Proximal Policy Optimization (PPO), and Advantage Actor-Critic (A2C). These algorithms represent different categories of DRL approaches and provide useful performance comparisons across varying traffic environments.

- **DQN** is a value-based method that approximates the Q-value function using a deep neural network. It selects actions by estimating the expected cumulative reward for each action in a given state and choosing the action with the highest value. DQN is suitable for discrete action spaces but may struggle with stability and sample efficiency in highly dynamic environments [31].
- **A2C** belongs to the actor-critic family, which combines a policy network (actor) for selecting actions and a value network (critic) for evaluating them. A2C enables better stability and learning efficiency by leveraging both value estimation and policy gradient updates. It performs well in moderate-sized state and action spaces and handles temporal credit assignment more effectively than DQN [40].
- **PPO** is a policy-based method that improves upon traditional policy gradient algorithms by using a clipped surrogate objective to ensure more stable and constrained updates. PPO balances exploration and exploitation effectively and is known for robustness and performance in continuous and discrete control problems. Its sample efficiency and stability make it well-suited for traffic control tasks involving complex environments [41].

These three algorithms were selected to represent different DRL families and to evaluate their relative performance across various traffic signal control setups.

To ensure a fair comparison, the policy, value, and action-value functions were designed with the same neural network architecture. We used a Python script to implement the DRL models via TraCI to accelerate deep learning. Tensor computations in the deep RL methods were handled by PyTorch [42] v1.11.0, and Stable-Baselines3 (SB3)

[43] v1.5.

4.5. Compared TSC methods

4.5.1. Fixed method

The fixed-time traffic signal control used in this study follows Webster's method, which calculates the optimal cycle time for minimizing delays at intersections. Webster's formula determines the cycle length based on the approach volumes, saturation flow rates, and the desired degree of saturation. The cycle length C is given by the formula:

$$C = \frac{1.5L + 5}{1 - Y} \tag{4}$$

Where, L represents the total lost time per cycle, including start-up delays and clearance times, while Y is the sum of the critical flow ratios for each approach. This method assumes stable traffic patterns and fixed signal timings across all time periods, without adapting to real-time traffic fluctuations.

4.5.2. Actuated method

In this study, an actuated traffic signal control system was used, which dynamically adjusts green phase durations based on real-time traffic detection. Vehicles are detected by loop detectors, and the green light is extended as long as vehicles are present within a certain time gap; otherwise, the signal transitions to yellow. The actuated system was calibrated in SUMO with the same green time settings as those for the DRL-based approaches, ensuring fair comparison.

4.6. Performance evaluation

Each model was executed for the same intersection 10 times using different random seeds to account for stochastic variations in traffic demand and driver behavior. This ensures that the results reflect robust average performance rather than being dependent on a specific traffic realization. Mean and standard deviation values are reported to capture performance variability. To ensure appropriate numerical precision, the reported values are rounded as follows: the number of conflicts (an integer-based metric) is presented with one decimal place; average delay and CO_2 emissions (continuous metrics) are presented with two decimal places. This approach accurately represents the statistical precision gained through averaging while aligning with the resolution of the simulation output.

5. Experiment results

This section presents the findings from our experiments for both synthetic and real scenarios.

5.1. Experiment results for synthetic scenarios

This subsection presents the results from simulations of synthetic traffic scenarios, used to evaluate algorithm performance. The set of possible actions is A = {North/South Green (NSG), East/West Green (EWG)}, with NSG allowing North-South travel and EWG for East-West travel. At each time step, the agent selects an action based on its strategy. The training involved three DRL approaches: value-based (DQN), policy-based (PPO), and actor-critic (A2C). For fairness, all algorithms used the same neural network architecture for policy, value, and action-value functions. Table 3 summarizes the parameters for each algorithm. Each agent was trained for 200 episodes (144,000 s), with 3600 time steps per episode, and all experiments used the same seed for consistency in traffic conditions and initial policies.

5.1.1. Impacts of state definitions

To evaluate the impact of different state definitions on TSC performance, we applied the three algorithms, PPO, A2C, and DQN, using Reward Function 1 as the baseline to train the agent across the four state definitions described in Section 4.3.

Table 3
Parameters used for training DRL algorithms.

Parameter	Value
Learning rate (α)	0.001
Discount factor	0.99
Minimum Batch size	128
n_epochs	10
Number of hidden layers	Two layers with 64 neurons each, using a ReLU activation function
Replay Memory size	50,000
Starting and ending exploration(ϵ)	1, 0.05
Optimizer	Adam

Figure 3 illustrates the performance of the agents with these various state definitions under the first traffic scenario. The results indicate that there is no significant difference in control performance when using different states. This suggests that low-resolution data from cameras is sufficient to ensure the effectiveness of DRL-based signal controllers, potentially eliminating the need for costly vehicular networks to collect precise vehicle data.

However, Fig. 3 also shows that State 4 requires more training time than the other states, indicating that this state definition demands greater computational resources and time to train the agent using the three different algorithms.

5.1.2. Impacts of action time interval

The action time interval is a crucial parameter for the model. It needs to be kept within reasonable limits, as both excessively long and short intervals can negatively impact model performance. We trained DRL models using three different time intervals,3 s, 5 s and 10 s, across the same number of episodes, with Reward Function 1 as the baseline under the first traffic scenario.

As shown in Fig. 4, the results indicate that 3-s and 5-s time interval allows all DRL agents to converge faster and more stably compared to using a 10-s interval.

5.1.3. Performance of DRL algorithms with different rewards for the 1st traffic scenario

Since performance across different state definitions was similar and State 4 had higher training complexity, we chose State 3 for further evaluation, as it offered better performance and lower training time. This state was used to assess the impact of different reward functions on algorithm performance.

Figure 5 shows the performance of PPO, A2C, and DQN with six reward functions in the first traffic scenario. It is evident from Fig. 5 that PPO converged faster and more stable than the other algorithms across all reward functions. DQN took longer to converge with Reward Functions 3, 4, and 5, while A2C underperformed with Reward Function 2.

Each model was evaluated for the same intersection 10 times using different random seeds to account for stochastic variations in traffic demand and driver behavior. This ensures that the results reflect robust average performance rather than being dependent on a specific traffic realization. Mean and standard deviation values are reported to capture performance variability. To avoid unrealistic traffic interactions that could result in extra simulated conflicts, a 100-m warm-up edge was added before each network entrance. The total simulation time was 100 min, with the first and last 15 min excluded, leaving 60 min for evaluation metrics such as average delay, conflicts, and CO2 emissions were calculated as shown in Table 4.DRL algorithms outperformed traditional controls, with delays between 8.062 and 8.370 s. DQN using Reward 2 and 5 achieved the lowest delays (8.062 and 8.273 s). Statistical analysis showed significant differences in delay ($p < 0.0001$). In terms of conflicts, the DRL algorithms had more than fixed method. PPO with

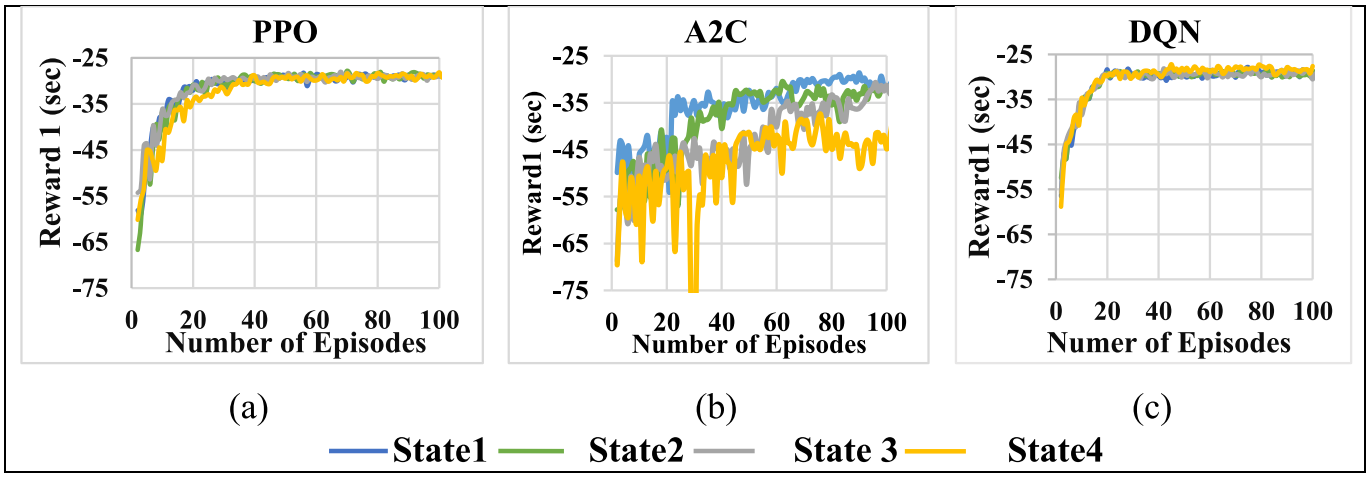


Fig. 3. Performance of DRL Agents with Varying State Definitions in First Scenario.

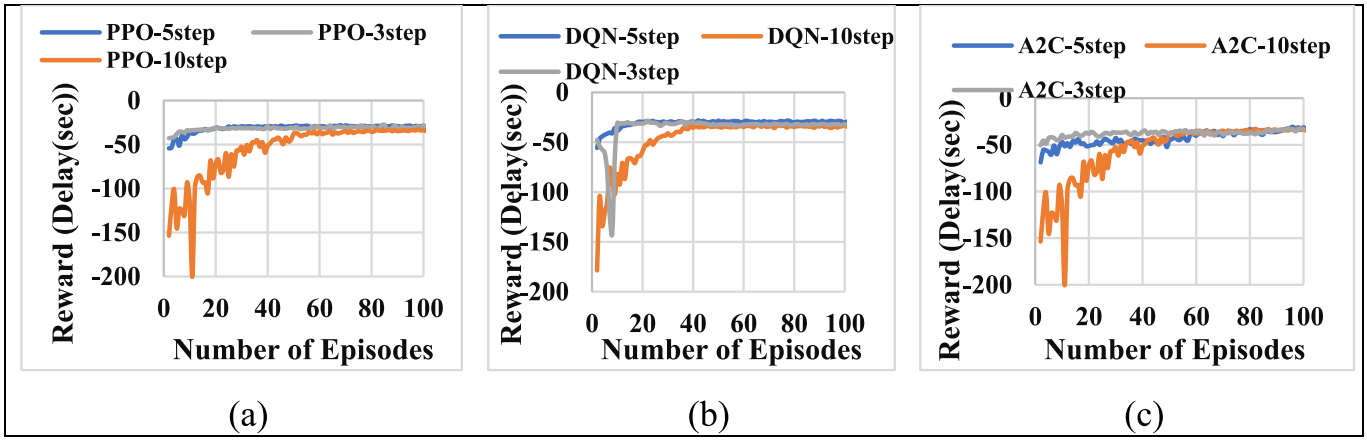


Fig. 4. Impact of Action Time Interval on the Performance of DRL Agents.

Reward 3 and Reward 5 showed the lowest conflict counts among DRL algorithms. For CO2 emissions, DRL algorithms generally outperformed traditional methods, with DQN using Reward 2 and 5 generating the lowest emissions (258,668.426 g and 289,527.122 g), with *p*-values below 0.0001 for statistical significance.

5.1.4. Relation between different rewards and traffic scenarios

The PPO algorithm was chosen for optimizing traffic signals for each traffic scenario due to its stability and faster convergence. The learned models were evaluated over 10 episodes, with metrics presented in Table 5 for all synthetic traffic scenarios.

Table 5 clearly shows that reward-based approaches generally outperformed traditional methods, showing lower average delays, fewer conflicts, and reduced CO_2 emissions. As traffic demand increased, the performance gap between reward-based and traditional methods widened, particularly in high-demand scenarios where reward-based methods significantly reduced conflicts. In low-volume scenarios, the fixed method had the fewest conflicts. However, Reward 4, designed to minimize CO_2 emissions, did not consistently reduce emissions, possibly due to external factors influencing emissions beyond the agent's control, This finding aligns with previous research, such as Schumacher et al. [13].

In Scenario 4, with the highest traffic demand, Reward 2 reduced average delay and conflicts by 22 % and 15 %, respectively, compared to the fixed method. Reward 3 showed even greater improvements, with a 7 % reduction in delay and a 21 % reduction in conflicts.

In terms of emissions reduction, while reward-based approaches offer incremental improvements, the differences remain relatively small in certain scenarios between DRL-based approaches and traditional methods. This could be attributed to that carbon emissions are closely tied to vehicle speed and acceleration profiles [44], which are not directly controlled by the DRL models in their current form. For instance, if a traffic signal allows for frequent stops or sharp accelerations, it can lead to higher emissions even if overall delay is reduced.

5.1.5. Sensitivity to multi- objective reward parameters

To assess the impact of different parameter weights in a multi-objective reward function on traffic signal optimization, the PPO algorithm was employed. This analysis focuses on Reward 5, where varying weights were assigned to traffic efficiency, safety, and CO_2 emissions. The sensitivity of the proposed models was evaluated under a variety of traffic demand conditions, including balanced and unbalanced traffic flows, and a wide range of volume-to-capacity (v/c) ratios, representing low, medium, and high demand levels. This comprehensive setup ensures that model performance is tested under both stable and highly congested scenarios.

Each weight configuration was trained for 100 episodes, followed by a 10-episode evaluation of the trained model.

Three parameter weight combinations were explored:

Equal weights efficiency, safety, and CO_2 emissions were each weighted equally

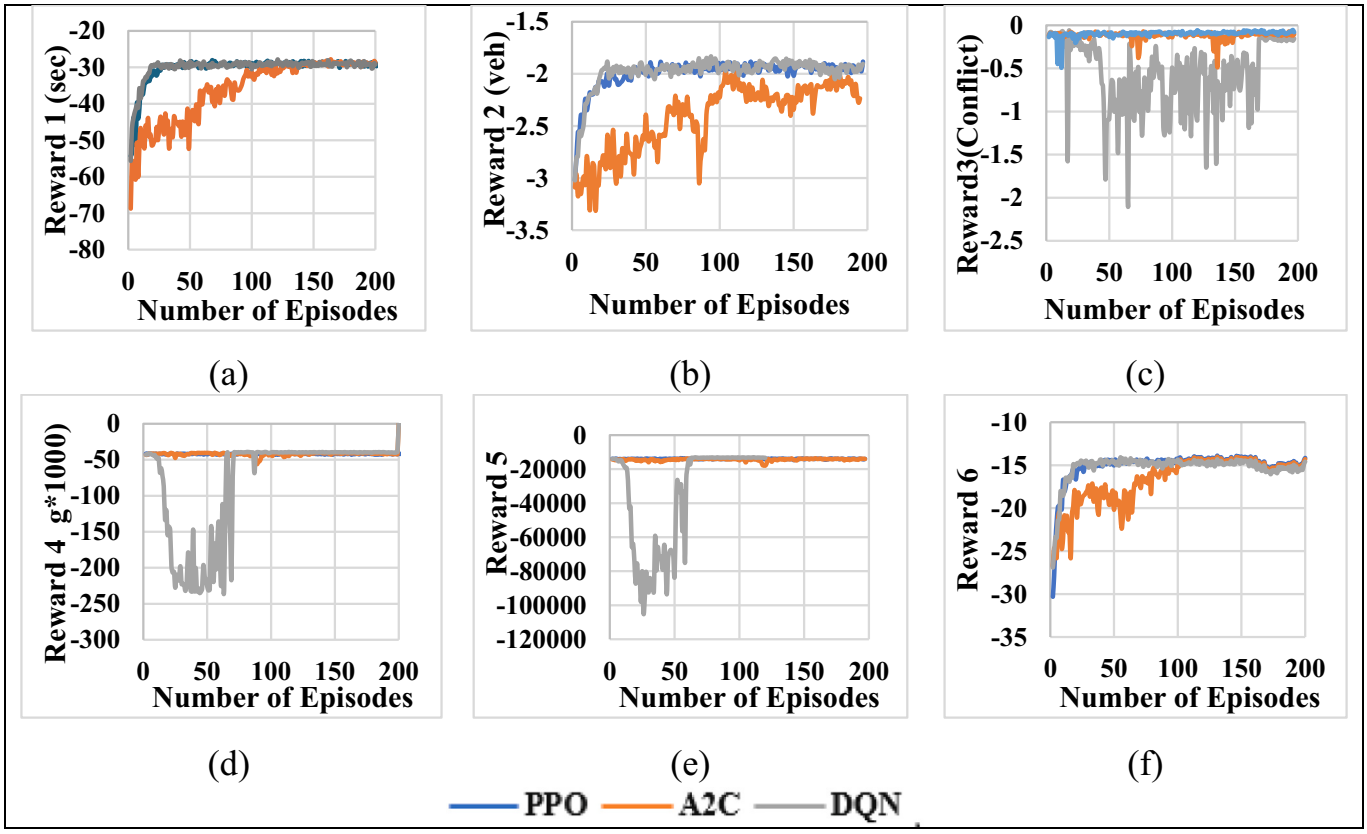


Fig. 5. Performance of DRL algorithms with different rewards for Fist Traffic Scenario (reward represents the average reward of the episodes).

$$Reward5-a = 0.33 \times Reward1(efficiency) + 0.33 \times Reward3(safety) + 0.33 \times Reward4(CO_2emissions)$$

Emphasis on efficiency and safety - 40 % was allocated to both efficiency and safety, and 20 % to CO_2 emissions

$$reward5-b = 0.4 \times Reward1(efficiency) + 0.4 \times Reward3(safety) + 0.2 \times Reward4(CO_2emissions)$$

Priority on efficiency - efficiency was weighted at 60 %, safety at 30 %, and CO_2 emissions at 10 %

$$reward5-c = 0.6 \times Reward1(efficiency) + 0.3 \times Reward3(safety) + 0.1 \times Reward4(CO_2emissions)$$

Table 6 presents detailed performance metrics for each scenario and reward configuration.

The sensitivity analysis shows that adjusting weights in the multi-objective reward function strongly affects traffic efficiency, safety, and CO_2 emissions. The efficiency-focused reward (Reward 5-c) minimizes delay and emissions but increases conflicts, potentially compromising safety. The balanced reward (Reward 5-a), with equal weights on all objectives, yields moderate performance across metrics, making it ideal for overall improvement. Meanwhile, the efficiency and safety-focused reward (Reward 5-b) reduces conflicts and maintains balanced delay and emissions, suitable when safety and efficiency are both priorities. This analysis underscores the flexibility of multi-objective optimization in tailoring traffic control to specific goals.

A deeper analysis of the results reveals several important trade-offs among mobility, safety, and environmental objectives. For instance, in high-demand conditions (e.g., Scenario 4 in Table 5), Reward 2, which emphasizes queue reduction, lowered average delays by 22 %, but increased traffic conflicts by 15 % compared to Reward 3, which is safety-focused. This reflects similar observations in [12]. Likewise, Reward 4, which prioritizes CO_2 emissions, sometimes led to longer delays or more conflicts, likely because of suboptimal phase switching that reduces vehicle stops but causes higher cumulative waiting time. As shown in Table 6, the efficiency-weighted configuration (Reward 5-c, with 60 % weight on delay) improved throughput by 18 % in Scenario 3 but raised CO_2 emissions by 12 % compared to the balanced configuration (Reward 5-a). This trade-off is consistent with findings in [13], which noted increased fuel consumption in acceleration-dominated stop-and-go regimes. These results confirm that aggressively optimizing a single objective can degrade others, emphasizing the importance of well-balanced reward functions in multi-objective DRL-based ATSC systems. The trade-offs also highlight the need for context-specific reward design. For example, peak-hour congestion may warrant the use of Reward 5-b (efficiency and safety weighted) to reduce delays while maintaining acceptable conflict levels, whereas school zones might prioritize Reward 3 to minimize conflicts, even if it results in modest increases in delay.

5.1.6. *PPO-based model training for a four-phase traffic scenario*

In this experiment, the PPO algorithm was trained using a four-phase traffic signal configuration to address increased left-turn demand, with left-turn movements set at 40 % and through and right-turn movements at 50 % and 10 %, respectively. This adjustment allowed for more precise left-turn control, reducing delays and managing conflicts in intersections with high left-turn volumes.

The model was trained for six reward functions over 100 episodes, followed by a 10-episode evaluation to measure the impact of the four-phase setup on key metrics: average delay, number of conflicts, and CO_2 emissions as shown in Table 7. It is shown from the table that the PPO model demonstrated adaptability to the complex four-phase configuration. Rewards 1, 2, and 6 achieved the lowest average delays (10.6–10.9 s), effectively optimizing traffic flow. Reward 3 resulted in the fewest conflicts (335.9), indicating improved safety. Rewards 2, 4, and 6

Table 4
Performance of evaluated episodes for the first traffic scenario.

Reward	Algorithm	Average Delay (sec)	Number of Conflicts	Total CO_2 emissions (gram)
Reward1	**PPO**	8.30 ± 0.08	128.1 ± 13.0	261,670.24 ± 975.79
	A2C	8.29 ± 0.09	128.1 ± 5.6	261,285.24 ± 904.55
	DQN	8.30 ± 0.08	128.2 ± 6.9	261,864.84 ± 1043.72
Reward2	**PPO**	8.34 ± 0.08	130.5 ± 8.7	261,595.96 ± 890.27
	A2C	8.37 ± 0.06	132.8 ± 8.3	262,269.33 ± 944.48
	DQN	**8.06 ± 0.12**	138.2 ± 8.1	**258,668.43 ± 1297.31**
Reward3	**PPO**	8.35 ± 0.11	123.4 ± 8.0	262,401.92 ± 1470.69
	A2C	8.33 ± 0.09	124. ±7.5	261,972.31 ± 760.63
	DQN	16.02 ± 0.20	122.4 ± 4.9	289,738.24 ± 1387.87
Reward4	**PPO**	8.31 ± 0.09	131.7 ± 12.4	262,218.32 ± 1143.27
	A2C	8.33 ± 0.10	127.9 ± 6.9	261,723.34 ± 1170.88
	DQN	8.28 ± 0.09	130.2 ± 9.3	261,872.24 ± 646.65
Reward5	**PPO**	8.29 ± 0.10	121.6 ± 10.0	261,679.75 ± 1257.48
	A2C	8.29 ± 0.10	127.4 ± 8.5	262,169.85 ± 1020.73
	DQN	16.02 ± 0.12	125.8 ± 4.9	289,527.12 ± 1798.48
Reward6	**PPO**	8.31 ± 0.12	123.9 ± 10.0	262,504.4 ± 1506.12
	A2C	8.36 ± 0.07	124.7 ± 10.3	262,145.50 ± 943.50
	DQN	8.30 ± 0.12	128.8 ± 7.7	262,087.19 ± 692.90
Fixed		13.41 ± 0.09	**104.0 ± 7.2**	268,408.89 ± 1388.30
Actuated		9.52 ± 0.08	131.6 ± 11.0	275,318.25 ± 2310.78

consistently minimized CO_2 emissions. In contrast, fixed-time and actuated controls produced significantly higher delays and emissions, reinforcing the efficiency of the PPO-based approach.

5.2. Experiment results for real-world intersections scenarios

To evaluate the proposed reward functions, models were trained for two real-world intersections described in Section 4.1. For the first intersection, the Deep Q-Network (DQN) model from Mei et al. [45] was utilized, applying their original reward function and the six proposed in this study. For the second intersection, DQN models from Mei et al. [45] and Ault and Sharon [41] were used with their respective reward functions, along with the six rewards developed in this research.

Each model underwent evaluation over 10 episodes with varied random seeds, measuring evaluation metrics as shown in Table 8, comparing with traditional methods. DRL-based models demonstrated superior performance in reducing delays, conflicts, and emissions across both intersections. For the first intersection, Reward 5 reduced delay by 6.6 % and CO_2 emissions by 2.2 % compared to the Mei et al. reward, while Reward 3 reduced conflicts by 5 %. At the second intersection, Reward 5 lowered delay by 10 % and 15 % and CO_2 emissions by 1.4 % and 6.4 % compared to the Mei et al. and Ault and Sharon rewards, respectively. Reward 3 reduced conflicts by 8 % and 14 % compared to these rewards.

Two models were calibrated for two intersections using the same state representation, delta time, and rewards adopted in this study.

Table 5
Performance of evaluated episodes across all synthetic traffic scenarios.

Traffic Scenario	Reward	Avg delay (sec)	Nnumber of conflicts	Total Co_2 emissions (gram)
Scenario 1	Reward 1	8.31 ± 0.08	128.1 ± 13.0	261,670.24 ± 975.79
	Reward 2	8.34 ± 0.08	130.5 ± 0.6	261,679.75 ± 1257.48
	Reward 3	8.35 ± 0.11	123.4 ± 8.0	262,401.92 ± 1470.69
	Reward 4	8.31 ± 0.09	131.7 ± 12.4	262,218.32 ± 1143.27
	Reward 5	**8.29 ± 0.10**	126.6 ± 10.0	**261,595.96 ± 890.27**
	Reward 6	8.31 ± 0.12	123.9 ± 10.0	262,504.48 ± 1506.12
	Fixed	13.41 ± 0.09	**104.0 ± 7.2**	268,408.89 ± 1388.30
	Actuated	9.52 ± 0.08	131.6 ± 11	275,318.25 ± 2310.78
Scenario 2	Reward 1	**9.92 ± 0.10**	203.7 ± 8.7	362,655.13 ± 1320.26
	Reward 2	9. 95 ± 0.06	196.3 ± 12.6	363,168.44 ± 1329.80
	Reward 3	10.20 ± 0.11	190.4 ± 12.1	363,348.44 ± 1776.21
	Reward 4	9.94 ± 0.10	197.2 ± 12.1	363,426.24 ± 1539.26
	Reward 5	9.931 ± 0.07	207.8 ± 12.9	**362,350.03 ± 1407.31**
	Reward 6	9.93 ± 0.10	203.3 ± 14.2	363,076.98 ± 1011.11
	Fixed	14.45 ± 0.11	**184.2 ± 12.1**	388,053.09 ± 2296.19
	Actuated	11.78 ± 0.150	174.3 ± 9.7	388,127.98 ± 1128.39
Scenario 3	Reward 1	**12.35 + 0.20**	298.9 ± 18.9	519,179.57 ± 3277.47
	Reward 2	12.52 ± 0.18	315.7 ± 16.3	521,048.12 ± 3388.50
	Reward 3	13.09 ± 0.17	**234.6 ± 17.0**	537,748.85 ± 2544.77
	Reward 4	19.23 ± 0.64	554.7 ± 62.4	579,845.00 ± 3637.90
	Reward 5	12.55 ± 0.33	308.0 ± 17.4	521,328.02 ± 4950.06
	Reward 6	12.37 ± 0.22	300.7 ± 17.8	**518,789.84 ± 2564.94**
	Fixed	15.89 ± 0.10	246.2 ± 13.7	532,602.24 ± 2196.69
	Actuated	18.48 ± 0.53	269.7 ± 16.6	577,249.33 ± 5065.92
Scenario 4	Reward 1	13.60 ± 0.33	259.2 ± 21.0	558,992.54 ± 3974.96
	Reward 2	**13.28 ± 0.07**	272.4 ± 15.9	**557,598.69 ± 2409.95**
	Reward 3	15.77 ± 0.15	**255.6 ± 30.2**	568,784.59 ± 3132.60
	Reward 4	29.23 ± 0.33	434.7 ± 52.5	688,458.62 ± 65,501.28
	Reward 5	16.79 ± 0.39	338.6 ± 16.5	593,824.36 ± 4532.61
	Reward 6	15.64 ± 0.28	318.9 ± 17.6	564,914.29 ± 3629.51
	Fixed	17.03 ± 0.12	323.1 ± 15.4	560,705.84 ± 5212.00
	Actuated	24.20 ± 1.21	391.1 ± 21.5	637,895.53 ± 6352.53
Scenario 5	Reward 1	**8.52 ± 0.09**	202.7 ± 13.3	237,054.37 ± 1067.51
	Reward 2	8.54 ± 0.07	199.5 ± 7.9	**236,265.13 ± 1198.20**
	Reward 3	8.55 ± 0.14	195.3 ± 10.9	237,154.33 ± 1324.06
	Reward 4	10.61 ± 0.39	198.4 ± 9.2	246,030.46 ± 3757.22

(*continued on next page*)

Table 5 (*continued*)

Traffic Scenario	Reward	Avg delay (sec)	Nnumber of conflicts	Total Co_2 emissions (gram)
Scenario 6	Reward 5	10.56 ± 0.47	196.0 ± 8.0	244,120.67 ± 3207.83
	Reward 6	8.55 ± 0.13	201.3 ± 11.3	236,782.06 ± 1238.88
	Fixed	12.21 ± 0.03	**76.0 ± 7.5**	246,794.57 ± 1071.27
	Actuated	8.57 ± 0.13	121.7 ± 15.6	250,539.48 ± 1789.89
	Reward 1	9.85 ± 0.09	147.4 ± 8.5	306,049.65 ± 1092.49
	Reward 2	**9.84 ± 0.12**	146.9 ± 10.0	306,249.41 ± 1245.60
	Reward 3	9.86 ± 0.12	**142.1 ± 15.1**	306,086.17 ± 1978.46
	Reward 4	9.86 ± 0.12	150.9 ± 12.8	**306,020.96 ± 1769.00**
	Reward 5	9.95 ± 0.13	147.8 ± 8.9	306,888.53 ± 1226.54
	Reward 6	9.86 ± 0.13	146.8 ± 9.5	306,887.46 ± 1316.43
	Fixed	12.22 ± 0.05	143.2 ± 13.9	320,740.89 ± 1218.75
	Actuated	9.61 ± 0.08	150.1 ± 11.0	323,454.93 ± 1817.99

Table 6
Performance of evaluated episodes across all synthetic traffic scenarios with varying parameter weightings within the multi-objective reward.

Traffic Scenario	Reward	Avg delay (*sec*)	Number of conflicts	Total Co_2 emissions (gram)
Scenario 1	Reward 1	8.29 ± 0.10	**121.6 ± 10.0**	**261,679.75 ± 1257.48**
	Reward 2	**8.28 ± 0.07**	124.0 ± 8.8	261,961.92 ± 1473.50
	Reward 3	8.34 ± 0.12	129.5 ± 6.6	262,121.69 ± 1185.01
Scenario 2	Reward 1	**9.93 ± 0.07**	207.8 ± 12.9	**362,350.03 ± 1407.31**
	Reward 2	9.97 ± 0.11	200.5 ± 6.5	362,593.55 ± 1407.30
	Reward 3	9.96 ± 0.16	**192.1 ± 8.2**	362,524.98 ± 1407.32
Scenario 3	Reward 1	12.55 ± 0.33	308.0 ± 17.4	521,328.02 ± 4950.06
	Reward 2	12.53 ± 0.20	307.0 ± 17.1	521,353.10 ± 4257.03
	Reward 3	**12.27 ± 0.22**	**287.4 ± 15.3**	**518,544.65 ± 3966.36**
Scenario 4	Reward 1	16.78 ± 0.38	338.6 ± 16.5	593,824.36 ± 4532.61
	Reward 2	16.54 ± 0.24	335.2 ± 10.4	593,913.36 ± 4253.06
	Reward 3	**14.65 ± 0.19**	**310.3 ± 9.9**	**591,824.54 ± 4432.94**
Scenario 5	Reward 1	10.56 ± 0.47	107.0 ± 8.0	244,120.67 ± 3207.83
	Reward 2	10.54 ± 0.53	106.1 ± 5.8	244,710.32 ± 2079.35
	Reward 3	**10.44 ± 0.35**	**105.3 ± 8.5**	**242,710.32 ± 2079.34**
Scenario 6	Reward 1	9.95 ± 0.13	147.8 ± 8.9	306,888.52 ± 1226.54
	Reward 2	9.94 ± 1.06	146.7 ± 18.5	306,515.86 ± 5381.87
	Reward 3	**9.88 ± 0.48**	**134.4 ± 26.9**	**302,352.86 ± 3107.81**

Some results from Table 5 are repeated here to facilitate direct comparison between different parameter weight settings in the multi-objective reward function (Reward 5). This helps illustrate how varying the emphasis on delay, conflicts, and emissions affects overall performance.

Table 7
Performance metrics for PPO model training under four-phase traffic signal configuration across different reward functions.

Reward	Avg delay(sec)	Number of conflicts	Total Co2 emissions(gram)
Reward 1	10.92 ± 0.18	344.3 ± 22.3	284,074.37 ± 1211.01
Reward 2	10.64 ± 0.14	374.3 ± 10.5	**283,187.83 ± 1337.65**
Reward 3	12.42 ± 0.30	**335.9 ± 16.8**	285,151.18 ± 1808.90
Reward 4	10.67 ± 0.15	376.9 ± 20.2	283,672.45 ± 943.65
Reward 5	10.67 ± 0.12	366.5 ± 22.5	284,371.67 ± 1068.16
Reward 6	**10.63 ± 0.15**	378.4 ± 10.3	283,582.17 ± 1042.32
Fixed	22.09 ± 0.20	374.6 ± 9.0	316,097.70 ± 1445.62
Actuated	22.20 ± 0.57	467.0 ± 14.3	327,303.01 ± 2228.00

Table 8
Performance of evaluated episodes for two real intersections.

#	Reward		Average Delay (sec)	Number of Conflicts	Total Co_2 emissions (gram)
First Intersection (Hangzhou, China)	Mei et al. Model reward		28.28 ± 0.48	1050.8 ± 17.1	186,757.26 ± 2452.06
	Mei et al. Model using Our rewards	**Reward1**	27.09 ± 0.81	1115.6 ± 24.5	183,029.35 ± 3816.90
		Reward2	26.90 ± 0.58	1175.4 ± 23.7	184,050.87 ± 3857.64
		Reward3	29.47 ± 0.93	**999.5 ± 18.9**	198,896.38 ± 5284.43
		Reward4	29.00 ± 0.72	1083.0 ± 17.5	194,390.94 ± 4735.71
		Reward5	**26.43 ± 0.46**	1064.5 ± 19.3	**182,570.26 ± 2884.11**
		Reward6	39.20 ± 1.02	1123.3 ± 22.8	239,632.27 ± 5960.63
	Fixed		203.15 ± 2.94	1341.5 ± 28.1	828,070.50 ± 11,212.43
	Actuated		48.19 ± 0.49	1283.6 ± 6.5	288,264.87 ± 2755.08
	Mei et al. Model		21.89 ± 0.51	1347.1 ± 28.4	158,862.60 ± 2165.09
	Ault and Sharon Model		22.95 ± 0.80	1440.7 ± 44.7	167,497.36 ± 3784.46
Second Intersection (Cologne, Germany)	Mei et al. Model using Our rewards	**Reward1**	21.45 ± 0.63	1387.8 ± 25.6	1,589,357.65 ± 2321.27
		Reward2	20.38 ± 0.50	1307.8 ± 27.5	158,227.96 ± 1888.56
		Reward3	20.55 ± 0.89	**1235.4 ± 34.9**	162,476.68 ± 5477.14
		Reward4	23.85 ± 0.96	1336.4 ± 43.6	168,423.30 ± 3148.50
		Reward5	**19.51 ± 0.58**	1277.6 ± 53.6	**156,712.54 ± 2011.49**
		Reward6	19.56 ± 0.42	1251.1 ± 13.4	157,853.11 ± 2585.14
	Fixed		96.21 ± 1.17	2318.3 ± 36.8	535,471.28 ± 6209.23
	Actuated		96.62 ± 1.14	2306.9 ± 30.8	538,015.31 ± 6970.71

Table 9 summarizes the parameters and model architecture used for both intersections. Table 10 highlights evaluation metrics, showing that models trained with Reward Functions 3, 5, and 6 consistently outperformed the Mei et al. model across all metrics. For the first intersection, Reward 5 achieved the lowest delay and emissions, while Reward 3 had the fewest conflicts. Similarly, at the second intersection, Reward 3 minimized delays and conflicts while maintaining low emissions, with all six reward functions surpassing the Mei et al. and Ault and Sharon models in delays and CO_2 emissions.

5.3. *Performance of multi-intersection scenario across different reward functions*

This study extends traffic signal control by applying DRL to a multi-

Table 9
Model parameters for the two real intersections.

Parameter	Intersection 1	Intersection 2
Algorithm used	DQN	
Learning rate (α)	0.001	0.005
Discount factor	0.95	
Minimum Batch size	64	256
Number of hidden layers	The model consists of four layers: one convolutional layer and three fully connected layers, each with 64 neurons and a ReLU activation function.	The model consists of two layers with 64 neurons in each one and a ReLU as activation function
Replay Memory size	5000	50,000
Starting and ending exploration(ϵ)	1.0,0.05	
Optimizer	RMSprop	Adam
Number of episodes	200	

Table 10
Performance of evaluated episodes at real intersections using our model.

Intersection	Reward	Average Delay (sec)	Number of Conflicts	Total Co_2 emissions (gram)
First Intersection (Hangzhou, China)	**Reward1**	24.58 ± 0.37	1151.9 ± 16.5	171,105.39 ± 2360.82
	Reward2	25.08 ± 0.91	1158.3 ± 24.6	171,342.09 ± 4131.53
	Reward3	24.28 ± 0.41	**979.6 ± 21.4**	170,198.16 ± 2573.11
	Reward4	27.22 ± 0.75	1114.5 ± 28.2	183,954.66 ± 4306.70
	Reward5	26.82 ± 0.87	1013.3 ± 19.2	183,549.03 ± 4764.37
	Reward6	**22.79 ± 0.34**	1025.1 ± 15.5	**164,332.50 ± 1909.45**
Second Intersection (Cologne, Germany)	**Reward1**	20.96 ± 0.78	1331.0 ± 47.3	163,517.05 ± 4178.16
	Reward2	19.4 ± 0.51	1257.4 ± 18.3	**152,132.86 ± 2097.05**
	Reward3	**19.00 ± 0.62**	**1226.0 ± 34.5**	153,090.33 ± 2521.54
	Reward4	21.91 ± 0.75	1344.1 ± 42.7	161,853.09 ± 6080.45
	Reward5	19.30 ± 0.24	1244.5 ± 30.4	154,250.49 ± 1773.57
	Reward6	19.74 ± 0.26	1226.0 ± 34.5	155,632.35 ± 1106.16

intersection scenario using both decentralized and centralized setups. In the decentralized approach, each intersection operates independently, optimizing mobility, safety, CO_2 emissions, and multi-objective trade-offs with six reward functions. After identifying the best-performing reward, its performance was compared to the centralized setup, which optimizes system-wide performance but faces scalability challenges as network size grows due to the complexity of observation spaces and action coordination.

Table 11
Training parameters for decentralized, and centralized setups.

Parameter	Value
Learning rate (α)	0.00025
Discount factor	0.99
Minimum Batch size	256
n_epochs	10
Number of hidden layers	Three layers with 64 neurons each, using a ReLU activation function
Optimizer	Adam
Number of training Episodes	100

Identical model parameters, detailed in Table 11, were used for consistency. The Proximal Policy Optimization (PPO) algorithm was employed to train models for the decentralized setup across six reward functions. Performance was evaluated over 10 episodes, with results in Table 12 showing that Reward 3 performed best, achieving the lowest average delay (25 s), the fewest conflicts (1522), and the lowest CO_2 emissions, effectively balancing mobility, safety, and environmental objectives. In contrast, Reward 1 had the highest delays and emissions, while Rewards 5 and 6 showed less consistent optimization.

Using the top-performing reward, Table 12 compares decentralized and centralized setups. The decentralized approach demonstrated superior performance, with a 26.4 % reduction in delays, a 26.9 % decrease in conflicts, and a 12.0 % reduction in CO_2 emissions compared to the centralized setup. These findings underscore the decentralized setup's ability to deliver precise control and better outcomes, making it ideal for managing large and complex traffic networks.

6. Summary and conclusions

This study leverages Deep Reinforcement Learning (DRL) to enhance adaptive traffic signal control (TSC) by calibrating models with reward functions that balance efficiency, safety, CO_2 emissions, and multi-objective goals. Three DRL algorithms (PPO, A2C, DQN) were tested against traditional TSC methods across synthetic traffic scenarios and validated at real-world intersections in Hangzhou and TAPAS Cologne. Key findings reveal that using low-resolution state data is effective and cost-efficient, and that shorter action intervals (3–5 s) improve convergence stability. In balanced, low-traffic scenarios, the PPO algorithm achieved up to 39 % delay reduction and 3.6 % lower CO_2 emissions compared to fixed methods. Reward functions significantly impacted DRL performance, with certain functions favoring safety or efficiency in high-traffic conditions. Additionally, modifying reward weights enabled adaptable TSC, balancing delays, safety, and emissions. In real-world tests, DRL models with custom rewards reduced delays, emissions, and conflicts compared to previous models. These findings highlight DRL's potential in TSC, though further research is needed to generalize results across diverse intersections and traffic patterns. Extending DRL to multi-intersection scenarios, the study compared decentralized and centralized setups. The decentralized approach, using six distinct reward functions, enabled intersections to operate independently. Among these, Reward 3 was the most effective, achieving the lowest delays, conflicts, and CO_2 emissions. In contrast, the centralized setup faced scalability challenges due to its unified observation space. The decentralized setup demonstrated superior performance, achieving 26.4 % reductions in delays, 26.9 % fewer conflicts, and 12 % lower CO_2 emissions compared

Table 12
Performance results of the overall system under different setup.

Agent	Reward	Avg delay (sec)	Number of conflicts	Total Co2 emissions (gram)
Decentralized	Reward 1	32.68 ± 0.84	2097.6 ± 84.1	595,467.01 ± 23,686.99
	Reward 2	29.63 ± 2.39	1542.6 ± 46.6	567,799.87 ± 26,858.86
	Reward 3	**25.54 ± 0.95**	**1522.3 ± 36.1**	**533,621.78 ± 7059.11**
	Reward 4	27.69 ± 0.97	1650.0 ± 44.3	552,759.79 ± 9876.71
	Reward 5	30.84 ± 3.28	2004.4 ± 91.5	564,675.57 ± 31,070.24
	Reward 6	30.19 ± 3.10	2028.1 ± 77.6	571,358.61 ± 11,534.76
Centralized	Reward3	34.70 ± 1.90	2082.4 ± 21.0	606,536.70 ± 13,507.23

to the centralized approach.

CRediT authorship contribution statement

Marwa Elharoun: Writing – review & editing, Writing – original draft, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sherif M. El-Badawy:** Writing – review & editing, Writing – original draft, Validation, Supervision, Software, Resources, Project administration, Conceptualization. **Elsayed Abd-Elazem Shwaly:** Writing – review & editing, Supervision, Methodology, Conceptualization. **Usama Elrawy Shahdah:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization.

Declaration of Competing interest

The authors declare no conflict of interest.

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